

Etienne Lanzeray

etienne.lanzeray.etu@univ-lille.fr
+33 6 98 29 96 42
Lille, France

GitHub
Website

INTERESTS	Mathematics and Machine Learning. Planning to pursue a PhD in ML approximations of PDE solutions. I'm also interested in 3D rendering (Overview article).
SKILLS	Data Science: Neural architectures for PDEs, PAC learnability and Rademacher bounds, Optimization, Bayesian Learning, NLP, Computer Vision, Reinforcement Learning, HPC, Fairness & Privacy. Mathematics: Distribution theory & PDEs, Spectral theory & Harmonic Analysis, Probabilities & Stochastic Processes, Statistics, Algebra, Analytic Number theory, Algebraic Topology & Differential Geometry, Complex Analysis. Programming: Python (Pandas, Scikit-learn, PyTorch, TensorFlow), Rust, SQL. Parallel & GPU: Open MPI, CUDA. Languages: French — native; English — TOEIC 970/990 (2025).
EDUCATION	Data Science, Master (<i>Research-oriented, English-taught</i>) <i>Université de Lille, Centrale Lille & IMT Nord Europe</i> Lille, France <i>M1 with highest honors (2024–2025)</i> <i>M2 (2025–2026)</i> Pure Mathematics, Master 1 <i>Magistère Jacques Hadamard¹</i> <i>Université Paris Saclay (2017–2018)</i> Orsay, France Pure and Applied Mathematics, B.Sc. <i>Magistère Jacques Hadamard</i> <i>Université Paris Saclay (2016–2017)</i> Orsay, France Classes préparatoires² in Mathematics, Physics and Computer Science <i>Lycée Gustave Eiffel (2013–2016)</i> Bordeaux, France
RESEARCH PROJECTS	Applying homogeneous Neural Networks to fluid mechanics models via homogeneous Galerkin projection 2025–2026 Advisors: Andrey Polyakov (Inria & CRISTAL, CNRS) Mihaly Petreczky (CRISTAL, CNRS) Performance evaluation and generalization bounds of homogeneous Neural Networks 2024–2025 Advisors: Andrey Polyakov (Inria & CRISTAL, CNRS) Mihaly Petreczky (CRISTAL, CNRS) (Report) The analytic large sieve and applications to number theory 2017–2018 Advisor: Etienne Fouvry (Institut de Mathématiques d'Orsay) (Report in french) Uniqueness of trigonometric series expansions 2016–2017 Advisor: Julien Sabin (Institut de Mathématiques d'Orsay)

¹Elite three-year honors program combining B.Sc. and M.Sc. in mathematics. Highly selective admission based on academic excellence (~30 students/year).

²Classes Préparatoires aux Grandes Écoles (CPGE): intensive post-secondary program unique to France, preparing students for competitive entrance examinations to elite engineering and science schools.

REFERENCES	Pierre Tirilly	Associate Professor at Université de Lille, Researcher at CRIStAL (CNRS), Head of the M1 Data Science program pierre.tirilly@univ-lille.fr
	Andrey Polyakov	Researcher at Inria Lille, Associate Researcher at CRIStAL (CNRS) andrey.polyakov@inria.fr
	Mihaly Petreczky	Researcher at CRIStAL (CNRS) mihaly.petreczky@centralelille.fr
RESEARCH EXPERIENCE	Research Intern	2025
	<i>Université de Lille</i>	
	Topic:	Performance evaluation and generalization bounds of homogeneous Neural Networks
	Advisors:	Andrey Polyakov (Inria & CRIStAL, CNRS) Mihaly Petreczky (CRIStAL, CNRS)
	Research Intern	2016
	<i>ICMCB (CNRS) & CEA Saclay</i>	
	Topic:	Mathematical modeling of the growth of nodules on silica nanoparticles and simulation implementation
	Advisor:	Etienne Duguet (ICMCB, CNRS)
TEACHING EXPERIENCE	Tutor in Computer Science (L1)	2024–2025
	<i>Université de Lille</i>	
	Topics:	Programming fundamentals, algorithms, basic data structures
	Description:	Support for first-year students during tutorials and labs
	Private Teacher	2019–2020
	<i>Academia</i>	
	Subjects:	Mathematics, physics–chemistry (secondary level)
	Oral Examiner (CPGE)	2017
	<i>Lycée Gustave Eiffel, Bordeaux</i>	
	Role:	Preparation of students for French engineering school entrance exams (30–45 min mathematics orals)
	Data Science Intern	2023–2024
	<i>Azur Drones, Mérignac</i>	
	Work:	Designed a flight database (TimescaleDB), built analytics for behavior and predictive maintenance, trained an LSTM autoencoder for anomaly detection on ~ 80 features ($> 90\%$ accuracy)
	Volunteer ML Collaborator	2023–2024
	<i>Ebiose (Inria Startup Studio)</i>	
	Project:	Contrastive-learning pipeline for selecting and scoring AI agents on <i>GSM8k</i>
	Repo:	github.com/EtiNL/Ebiose_CLGP
	Student Association Roles	2016–2018
	<i>Université Paris Saclay</i>	
	Roles:	Event manager for the Institut Mathématique d’Orsay (BDE Math $\sim$) Network and activities manager of the CROUS residence association (Eminet)
HOBBIES	Backpacking and multi-day high-mountain trekking, Skateboard, Chess & Guitar	